

NONCENTRAL MATRIX T DISTRIBUTIONS

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SUMMARY

Two noncentral forms of the matrix T distribution are defined: the lower non-central and the upper noncentral distributions. It is shown that, in certain cases, the matrix of sample regression coefficients from a multivariate normal population follows a lower noncentral T distribution. This result has applications to the distribution of two-stage least squares estimators.

Key Words: Multivariate t distributions, two-stage least squares estimators, regression analysis

1. Introduction

The central matrix T distribution was defined by Dickey (1967) as the distribution of the $p \times q$ matrix

$$(1) \quad T = (U^{\frac{1}{2}})^{-1}X$$

where U is a $p \times p$ matrix having a central Wishart distribution with covariance matrix $P > 0$ and $(m-q)$ degrees of freedom ($U \sim W_p(P, m-q)$), X is a $p \times q$ matrix the row vectors of which are independent, each having a multivariate normal distribution with zero mean and covariance matrix $Q > 0$; U and X are independent and $U^{\frac{1}{2}}$ is the symmetric square root of U . The density function of T is proportional to $|P|^{-1} + |TQ^{-1}T'|^{-\frac{1}{2}m}$. Density functions of this form had appeared earlier in the literature, firstly in connection with classical multivariate normal regression, Kshirsagar (1960), and later in a Bayesian context, Tiao and Zellner (1964) and Geisser (1965).

In this paper we define two noncentral matrix T distributions, the lower non-central matrix T in which U has a noncentral Wishart distribution and the upper noncentral matrix T distribution in which the rows of X are independent normal vectors with non-zero means. We have not succeeded in expressing the density function of the upper noncentral T in closed form. In the final section we show that the lower noncentral matrix T has application in regression analysis.

2. Preliminary Results

$N_q(\mu, Q)$ denotes a q -dimensional normal distribution with mean vector μ and covariance matrix Q . $W_p(P, m-q)$ denotes the central Wishart distribution with covariance matrix P and $(m-q)$ degrees of freedom, $W_p(P, m-q, \Omega)$ denotes the noncentral Wishart distribution where $(\omega_{ij}) \equiv \Omega$ is the noncentrality parameter. We consider also the canonical form of the linear noncentral Wishart distribution, Anderson (1946). In this case P is the identity matrix and the only non-zero element of Ω is $\omega_{11} = \kappa^2$. We denote this distribution by $W_p(I_p, n, \kappa^2)$.

3. The lower noncentral matrix T distribution

Theorem 1

Suppose that the row vectors of the pxq matrix X are each independently distributed as $N_q(O, Q)$, $Q > O$ and $U \sim W_p(P, m-q, \Omega)$, $P > O$ and $m > p+q-1$, independently of X . If $U^{1/2}$ is the symmetric square root of U , then the density function of

$$T = (U^{1/2})^{-1}X$$

is

$$\frac{\Gamma_p(\frac{1}{2}m) \text{etr}(-\Omega) |P|^{-1} + TQ^{-1}T'|^{-\frac{1}{2}m}}{\pi^{\frac{1}{2}pq} \Gamma_p(\frac{1}{2}(m-q)) |P|^{\frac{1}{2}(m-q)} |Q|^{\frac{1}{2}p}} \cdot {}_1F_1(a; b; c)$$

with $a = \frac{1}{2}m$, $b = \frac{1}{2}(m-q)$ and $c = P^{-1}\Omega(P^{-1} + TQ^{-1}T')^{-1}$.

Proof

The density function of X is

$$f_1(X) = \frac{\text{etr}(-\frac{1}{2}Q^{-1}X'X)}{(2\pi)^{\frac{1}{2}pq} |Q|^{\frac{1}{2}p}}$$

and

$$f_2(U) = \frac{\text{etr}(-\Omega) |U|^{\frac{1}{2}(m-q-p-1)} \text{etr}(-\frac{1}{2}P^{-1}U)}{2^{\frac{1}{2}(m-q)} \Gamma_p(\frac{1}{2}(m-q)) |P|^{\frac{1}{2}(m-q)}} \cdot {}_0F_1(\frac{1}{2}(m-q), \frac{1}{2}P^{-1}\Omega U).$$

Expanding the ${}_0F_1$ series in terms of zonal polynomials, $C_\kappa(\cdot)$, gives

$$f_2(U) \propto |U|^{\frac{1}{2}(m-q-p-1)} \text{etr}(-\frac{1}{2}P^{-1}U) \sum_{\kappa=0}^{\infty} \sum_{\kappa} \frac{C_\kappa(\frac{1}{2}P^{-1}\Omega U)}{(\frac{1}{2}(m-q))_\kappa k!}$$

The joint density of X and U is

$$f_3(X, U) \propto |U|^{\frac{1}{2}m-q-p-1} \text{etr}(-\frac{1}{2}(P^{-1}U + XQ^{-1}X')) \sum_{\kappa=0}^{\infty} \sum_{\kappa} \frac{C_\kappa(\frac{1}{2}P^{-1}\Omega U)}{(\frac{1}{2}(m-q))_\kappa k!}$$

where the constant of proportionality is

$$(2^{\frac{1}{2}mp} \pi^{\frac{1}{2}pq} \Gamma_p(\frac{1}{2}(m-q)) |P|^{\frac{1}{2}(m-q)} |Q|^{\frac{1}{2}p})^{-1}$$

Transforming in $f_3(X, U)$ by

$X = (U^{\frac{1}{2}})T$ with $J(X \rightarrow T) = |U|^{\frac{1}{2}q}$, where $J(X \rightarrow T)$ is the Jacobian of the transformation, gives

$$f_3(T, U) \propto |U|^{\frac{1}{2}(m-p-1)} \text{etr}(-\frac{1}{2}(P^{-1} + TQ^{-1}T')U) \cdot \sum_{k=0}^{\infty} \frac{C_k(\frac{1}{2}P^{-1}\Omega U)}{(\frac{1}{2}(m-q))_k k!}$$

Integrating termwise over U using a result of Constantine (1963), p. 1271, with $R = \frac{1}{2}(P^{-1} + TQ^{-1}T')$ and $W = \frac{1}{2}P^{-1}\Omega$ (where W denotes Constantine's T), gives

$$f_4(T) \propto 2^{\frac{1}{2}mp} |P^{-1} + TQ^{-1}T'|^{-\frac{1}{2}m} \sum_{k=0}^{\infty} \frac{\Gamma_p(\frac{1}{2}m, \kappa) C_k(c)}{(\frac{1}{2}(m-q))_k k!}$$

where $\Gamma_p(\frac{1}{2}m, \kappa)$ denotes the multivariate gamma function, and $c = P^{-1}\Omega(P^{-1} + TQ^{-1}T')^{-1}$. Using the result (James, 1964)

$$\Gamma_p(\frac{1}{2}m, \kappa) = (\frac{1}{2}m)_\kappa \Gamma_p(\frac{1}{2}m)$$

the infinite series can be expressed as $\Gamma_p(\frac{1}{2}m)_1 F_1(a; b; c)$. Evaluation of the constant term gives the density function stated in the theorem.

We denote the lower noncentral T distribution as defined in Theorem 1 by $T(P, Q, 0, m, \Omega)$, the zero referring to the fact that the rows of X have zero mean and m is the number of degrees of freedom. It is easily shown that any columnwise partition of T is also distributed as lower noncentral T .

The linear case of the lower noncentral T arises when the rows of X are $N_q(O, I_q)$ and $U \sim W_p(I_q, m-q, \kappa^2)$. The derivation follows along the lines of Theorem 1 and the density function is

$$(2) \quad \frac{\exp(-\frac{1}{2}\kappa^2) \Gamma_p(\frac{1}{2}m)}{\pi^{\frac{1}{2}pq} \Gamma_p(\frac{1}{2}(m-q))} |I_p + TT'|^{-\frac{1}{2}m} {}_1F_1(\frac{1}{2}m; \frac{1}{2}(m-q); \frac{1}{2}\kappa^2 T^{-1})$$

where κ^2 is the scalar noncentrality parameter and $(\tau^{jj}) \equiv (I_p + TT')^{-1}$. Application of Lemma 13.3.1 of Anderson (1958), page 319, shows that in this instance, $V = TT'$ has a noncentral multivariate Beta Type 2A distribution (cf. De Waal (1968)).

4. The Upper Noncentral Matrix T distribution

To define the upper noncentral matrix T distribution we assume that the row vectors of X are independently distributed as $N_q(\mu^{(i)}, Q)$ $i = 1, \dots, p$; U has a central $W_p(P, m-q)$ distribution and is independent of X . In this case it does not seem possible to express the density of $T = (U^{1/2})^{-1}X$ in closed form. If μ is the $p \times q$ matrix whose row vectors are the expected values of the rows of X , then it is easily shown that to obtain $f_4(T)$ we need to find

$$\int_{U>0} |U|^{1/2(m-p-1)} \text{etr}(-1/2(P^{-1} + TQ^{-1}T')U) \text{etr}(U^{1/2}'TQ^{-1}\mu) dU$$

This integral appears to be difficult to evaluate.

5. The distribution of $B = A_{11}^{-1}A_{12}$ when A has a noncentral Wishart distribution (linear case)

If A has a central Wishart distribution with covariance matrix Σ and n degrees of freedom, $A \sim W(\Sigma, n)$, the matrix of sample regression coefficients, $B = A_{11}^{-1}A_{12}$ has a matrix T distribution (cf. Kshirsagar (1960), Kabe (1968) and Dickey (1967)). We show that if A has a linear noncentral Wishart distribution $A \sim W(I, n, \kappa^2)$, the distribution of B is lower noncentral matrix T. It can be shown that the two-stage least squares estimates of the coefficients of the endogenous variables in a system of linear structural equations can be represented as B . Special cases of their distribution have been given to the literature by Basman (1961, 1963), Kabe (1963, 1964), Richardson (1968) and Sawa (1969), but the connection between these and the lower noncentral matrix T does not appear to have been recognised. A full discussion of this is given in Juritz (1973). The distribution of B when A has a noncentral Wishart distribution of any rank appears to be intractable.

Theorem 2

Let A have a linear noncentral Wishart distribution with covariance matrix I_p , n degrees of freedom and noncentrality parameter κ^2 , $A \sim W(I_p, n, \kappa^2)$, and let A be partitioned as $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ where A_{11} is $q \times q$, A_{12} is $q \times r$, A_{22} is $r \times r$.

If $B = A_{11}^{-1}A_{12}$ then

$$(3) \quad f_5(B) = \frac{\exp(-1/2\kappa^2) \Gamma_q(1/2(n+r))}{\pi^{1/2} q^r \Gamma_q(1/2n)} |I_q + BB'|^{-1/2(n+r)} \cdot {}_1F_1(1/2(n+r); 1/2n; 1/2\kappa^2 \omega^{11})$$

where $(\omega^{ij}) \equiv (I_q + BB')^{-1}$.

Proof

The density function of A is

$$f_6(A) \propto |A|^{\frac{1}{2}(n-p-1)} \text{etr}(-\frac{1}{2}A) \sum_{i=0}^{\infty} \phi(i) a_{11}^i$$

where the constant of proportionality is

$$(4) \quad \exp(-\frac{1}{2}\kappa^2) (2^{\frac{1}{2}np} \Gamma_p(\frac{1}{2}n))^{-1}$$

and

$$(5) \quad \phi(i) = (\frac{1}{2}\kappa^2)^i \Gamma(\frac{1}{2}n)(i! 2^i \Gamma(\frac{1}{2}n+i))^{-1}$$

Writing $f_6(A)$ in terms of the partitioning above we have

$$f_7(A_{11}, A_{12}, A_{22}) \propto |A_{11}|^{\frac{1}{2}(n-p-1)} |A_{22} - A_{21} A_{11}^{-1} A_{12}|^{\frac{1}{2}(n-p-1)} \\ \cdot \text{etr}(-\frac{1}{2}A_{11}) \text{etr}(-\frac{1}{2}A_{22}) \sum_{i=0}^{\infty} \phi(i) a_{11}^i$$

Transforming from A_{12} to B by $B = A_{11}^{-1} A_{12}$ with $J(A_{12} \rightarrow B) = |A_{11}|^r$ and noting $p = q+r$, we find

$$f_8(A_{11}, A_{22}, B) \propto |A_{11}|^{\frac{1}{2}(n+r-q-1)} |A_{22} - B' A_{11} B|^{\frac{1}{2}(n-q-r-1)} \\ \cdot \text{etr}(-\frac{1}{2}A_{11}) \text{etr}(-\frac{1}{2}A_{22}) \sum_{i=0}^{\infty} \phi(i) a_{11}^i.$$

We first integrate over A_{22} and require

$$\int |A_{22} - B' A_{11} B|^{\frac{1}{2}(n-q-r-1)} \text{etr}(-\frac{1}{2}A_{22}) dA_{22}$$

Letting $A_{22} - B' A_{11} B = W$ with $J(A_{22} \rightarrow W) = 1$, the integral becomes

$$\text{etr}(-\frac{1}{2}B' A_{11} B) \int_{W > 0} |W|^{\frac{1}{2}(n-q-r-1)} \text{etr}(-\frac{1}{2}W) dW$$

which has the value

$$(6) \quad \text{etr}(-\frac{1}{2}B' A_{11} B) \Gamma_r(\frac{1}{2}(n-q)) 2^{\frac{1}{2}r} \pi^{r(n-q)}.$$

Hence

$$f_9(A_{11}, B) \propto |A_{11}|^{1/2(n+r-q-1)} \text{etr}(-1/2(I_q + BB')A_{11}) \sum_{i=0}^{\infty} \phi(i) a_{11}^i.$$

We must now integrate termwise over A_{11} and must evaluate integrals of the form

$$\int a_{11}^i |A_{11}|^{1/2(n+r-q-1)} \text{etr}(-1/2(I_q + BB')A_{11}) dA_{11}.$$

This integral is proportional to $E(a_{11}^i)$ where $(a_{ij}) \equiv A_{11} \sim W_q(n+r, (I_q + BB')^{-1})$. The value of the integral is

$$(7) \quad \frac{2^{1/2(n+r)} q \Gamma_q(1/2(n+r))}{|I_q + BB'|^{1/2(n+r)}} \cdot \frac{2^i \Gamma(1/2(n+r+i)) (\omega^{11})^i}{\Gamma(1/2(n+r))}$$

where $(\omega^{ij}) \equiv (I_q + BB')^{-1}$.

Evaluating the constant terms from (4), (5), (6) and (7) gives

$$(8) \quad f_5(B) = \frac{\exp(-1/2 \kappa^2) \Gamma_r(1/2(n-q)) \Gamma_q(1/2(n+r))}{\Gamma_p(1/2 n) |I_q + BB'|^{1/2(n+r)}} \cdot \sum_{i=0}^{\infty} \frac{(1/2 \kappa^2)^i \Gamma(1/2 n) \Gamma(1/2(n+r+i)) (\omega^{11})^i}{i! \Gamma(1/2(n+i)) \Gamma(1/2(n+r))}$$

The infinite series in (8) is ${}_1F_1(1/2(n+r); 1/2 n; 1/2 \kappa^2 \omega^{11})$ and

$$\frac{\Gamma_r(1/2(n-q))}{\Gamma_p(1/2 n)} = \frac{1}{\pi^{1/2} q \Gamma_q(1/2 n)}.$$

Hence we obtain the density function of B as given in (3). Comparison of (3) with (2) shows that the distribution of B is lower noncentral matrix T with $(n+r)$ degrees of freedom and noncentrality parameter κ^2 .

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